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Financing Competitors: Shadow Banks' Funding and Mortgage Market Competition

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1 Big picture

- **Question.** How are shadow banks funded, and what does that funding structure imply for competition in the mortgage market?
- **Setting.** The paper studies the U.S. residential mortgage market, where shadow banks originate loans but rely heavily on short-term warehouse lines of credit to finance the period between origination and sale.
- **Core challenge.** Shadow banks look like competitors to banks in the downstream mortgage origination market, but their balance-sheet model makes them dependent on upstream funding providers. If those providers are the same banks they compete with, then competition in mortgage origination may be shaped by market power in the funding market.
- **Main empirical result.** Shadow banks are predominantly funded by the very banks with which they compete in mortgage origination. Funding relationships are concentrated among lenders and borrowers with high downstream mortgage-market overlap, and those relationships are persistent and not easily substituted away.
- **Main mechanism.** Banks appear to have *upstream market power* in warehouse lending because warehouse lines are differentiated from the perspective of shadow banks. That market power softens downstream competition through two channels:
 - the **cannibalization channel**: banks charge higher warehouse rates to protect downstream mortgage profits from shadow-bank competition;
 - the **funding channel**: because banks earn upstream profits from lending to shadow banks, they internalize downstream competition and keep their own mortgage rates high.
- **Main quantitative result.** In the calibrated model, banks' upstream market power raises shadow banks' funding costs by about 12.4%, increases their mortgage rates, and reduces mortgage credit supply and consumer surplus.
- **Distributional takeaway.** The effects are stronger in more concentrated mortgage markets and in places where households value shadow-bank services more. So the households who would benefit most from shadow-bank competition are exactly those who bear the largest costs of banks' upstream market power.
- **Policy result.** Secondary-market innovations—such as eMortgage adoption and GSE reforms that shorten warehouse duration—can materially weaken banks' upstream market power and improve downstream mortgage competition.
- **Why this paper matters.** The paper links industrial organization and household finance: it shows that the organization of credit supply upstream changes prices, quantities, and welfare downstream. The bank-versus-shadow-bank margin is therefore not enough; one must also understand *who funds the shadow banks*.

2 Data and measurement (key objects)

2.1 Observed shadow-bank funding panel

- **Main innovation.** The paper constructs a novel loan-level dataset on warehouse funding relationships using shadow-bank call reports required under the S.A.F.E. Mortgage Licensing Act.
- **Coverage.** The raw call-report collection is obtained through FOIA requests from Washington and Massachusetts. Because the filings record nationwide operations for firms licensed in those states, the sample still captures about 80% of total shadow-bank mortgage origination in the United States.
- **Frequency.** Funding data are quarterly from 2011 to 2017.
- **Objects observed on each credit line.** The call reports identify the warehouse funding provider, credit limit, remaining balance / usage, and information that allows the paper to estimate funding costs.
- **Raw scale.** The paper identifies more than 200,000 funding relationships between 544 shadow banks and 399 funding providers. In the HMDA-matched estimation sample summarized in Table 1, there are 416 shadow banks, 207 warehouse banks, and 272,543 bank-shadow-bank pairs.
- **Balance-sheet relevance.** About 90% of shadow banks' mortgage assets on balance sheet are financed by warehouse credit.

2.2 Observed mortgage origination and bank-level data

- **Mortgage origination data.** The paper links warehouse funding relationships to HMDA loan-level mortgage origination data, which provide near-universe information on mortgage originations.
- **Bank data.** Funding providers are matched to regulatory call reports (Forms 031 and 041) and the Summary of Deposits, which provide bank balance sheets, income statements, and branch networks.
- **Additional loan-level data.** For mortgage-rate and borrower-risk analysis, the paper also uses conforming-mortgage disclosure data from the GSEs and Ginnie Mae.
- **Geography.** The reduced-form funding relationship analysis uses county-level and bank-shadow-bank-pair variation. The mortgage pricing analysis is conducted at the MSA level. The model calibration is at the MSA-year level.

2.3 Key measured objects

- **Mortgage market overlap.** For bank-shadow-bank pair (i, j) , mortgage market overlap is the share of counties in which both originate mortgages relative to the counties in which either operates:

$$\text{Overlap}_{ijt} = \frac{\sum_k I(\sigma_{ikt} > 0, \sigma_{jkt} > 0)}{\sum_k I(\sigma_{ikt} > 0) + \sum_k I(\sigma_{jkt} > 0)}. \quad (2)$$

- **Downstream concentration.** Mortgage market competition is measured by the county-level Herfindahl-Hirschman index (HHI).
- **Warehouse lending market share.** For a bank in an MSA-year, this is the share of shadow banks in that market financed by the bank.
- **Funding cost measures.** The paper estimates warehouse interest-rate spreads and uses a differential-net-worth shock to banks as a source of exogenous funding-cost variation.
- **Choice-set proximity.** The paper measures the minimum distance from a shadow bank headquarters to a bank branch network, and uses this both as a control and in robustness exercises that restrict shadow-bank lender choice sets.

2.4 Unobservables and timing

- **Warehouse relationship value.** In the model, the key unobservable is pairwise convenience ξ_{bs} , which captures the extra value a shadow bank gets from borrowing from bank b . It can be interpreted as relationship lending, financing flexibility, or another form of differentiated service.
- **Household preference shocks.** Mortgage demand contains idiosyncratic preference shocks ε_{ij} over lender options.
- **Timing in the model.** The stage game is:
 - banks set warehouse rates,
 - shadow banks choose warehouse line providers,
 - banks and shadow banks set mortgage rates,
 - households choose among mortgage products and an outside option,
 - shadow banks draw on warehouse lines and originate loans.
- **Warehouse duration.** In the data and model, warehouse financing bridges the period between origination and sale, typically about one to three months.

3 Models and empirical specifications (all equations + interpretation)

3.1 Matching in the warehouse lending market

The baseline matching regression is

$$\Pr(\text{Lend}_{ijt}) = \beta \text{Overlap}_{ijt} + \gamma \ln(\text{HQDistance}_{ijt}) + \mu_{it} + \mu_{jt} + \varepsilon_{ijt}. \quad (1)$$

- **Outcome.** Lend_{ijt} equals 100 if shadow bank i borrows from bank j in year t , and 0 otherwise.
- **Interpretation.** The coefficient on overlap asks whether bank-shadow-bank pairs with greater downstream mortgage overlap are more likely to form upstream funding relationships.
- **Identification logic.** Shadow-bank-year and bank-year fixed effects absorb time-varying borrower-side and lender-side factors; identification comes from cross-pair differences in overlap within those sets.

- **Main finding.** A one-standard-deviation increase in overlap raises the probability of a funding relationship by about 1.88 percentage points, which is economically large relative to the unconditional financing probability of 2.77%.

3.2 Testing substitutability of warehouse lenders

To test whether warehouse lenders are close substitutes, the paper exploits the 2014 oil-price collapse as a net-worth shock to banks with high exposure to the oil and gas sector. The main difference-in-differences specifications are

$$r_{it} = \beta \text{Treat}_i \times \text{Post}_t + \mu_i + \mu_t + \varepsilon_{it}, \quad (3a)$$

$$\text{Pr}(\text{Lend}_{ijt}) = \beta \text{HighOilShock}_j \times \text{Post}_t + \mu_{it} + \nu_j + \varepsilon_{ijt}, \quad (3b)$$

$$\text{Usage}_{ijt} = \beta \text{HighOilShock}_j \times \text{Post}_t + \mu_{it} + \nu_{ij} + \varepsilon_{ijt}. \quad (3c)$$

- **Equation (3a).** Treated shadow banks are those relying heavily on banks exposed to oil-and-gas losses. The outcome is funding cost.
- **Equations (3b)–(3c).** The first outcome is the probability of borrowing from a given bank, and the second is realized usage of a warehouse line.
- **Interpretation.** If warehouse lenders were perfect substitutes, treated shadow banks should switch to unexposed banks when exposed banks raise rates.
- **Main finding.** Treated shadow banks' funding costs rise by over 60 bps, but they do *not* switch providers in a statistically meaningful way. Instead, they reduce usage of exposed lines. This is direct evidence that warehouse lenders are differentiated.

3.3 The cannibalization channel in reduced form

The paper tests whether banks finance fewer shadow banks in less competitive local mortgage markets:

$$s_{jkt}^w = \beta \text{HHI}_{kt} + X_{kt}\Gamma + \mu_{jt} + \nu_{st} + \varepsilon_{jkt}. \quad (4)$$

- **Outcome.** s_{jkt}^w is the share of shadow banks in county k and year t financed by bank j .
- **Interpretation.** If downstream profits are more valuable in concentrated markets, a bank that also originates mortgages should be less willing to fund local shadow-bank competitors there.
- **Main finding.** A one-standard-deviation increase in downstream HHI is associated with a 58-bps decline in the share of shadow banks financed by a local bank.

3.4 The funding channel in reduced form

The paper then studies whether banks that fund more shadow banks charge higher mortgage rates themselves:

$$r_{ljk} = \alpha + \beta s_{jkt}^w + X_l\Phi + \delta_{jt} + \mu_{kt} + \varepsilon_{ljk}. \quad (5)$$

- **Outcome.** r_{ljk} is the mortgage rate on loan l originated by bank j in MSA k at time t .

- **Key regressor.** s_{jkt}^w is the share of shadow banks in that MSA-year financed by bank j .
- **Interpretation.** A bank with a larger stake in shadow-bank profits internalizes more of the downstream competition and therefore softens its own pricing.
- **Main finding.** A one-percentage-point increase in warehouse lending market share is associated with about a 19-bps increase in the mortgage rate the bank charges.

3.5 Bank problem: upstream warehouse pricing

In stage 1, a bank chooses its warehouse lending rate to maximize total profit from upstream warehouse lending and downstream mortgage origination:

$$\max_{\rho_b} \eta(\rho_b - \psi_b^w) \sum_{s \in S} s_{sb}^w(\rho_b) d_s^*(\rho_b) + \pi_b^*(\rho_b). \quad (6)$$

The corresponding optimal pricing rule is

$$\rho_b = \psi_b^w - \frac{\eta \sum_{s \in S} s_{sb}^w(\rho_b) d_s^*(\rho_b)}{\Delta} - \frac{\partial \pi_b^*(\rho_b)}{\partial \rho_b} \Delta^{-1}, \quad (7)$$

with

$$\Delta \equiv \eta \sum_{s \in S} \frac{\partial s_{sb}^w(\rho_b)}{\partial \rho_b} d_s^*(\rho_b) + \sum_{s \in S} s_{sb}^w(\rho_b) \frac{\partial d_s^*(\rho_b)}{\partial \rho_b}.$$

- **Interpretation of (6).** A vertically integrated bank cares about both warehouse-lending profits and the effect of warehouse pricing on downstream mortgage profits.
- **Interpretation of (7).** The upstream price contains a baseline markup plus an additional markup that reflects the incentive to raise shadow-bank funding costs in order to protect downstream mortgage profits.
- **Economic meaning.** This additional markup is the model counterpart of the **cannibalization channel**.

3.6 Bank problem: downstream mortgage pricing

Given warehouse relationships, bank b sets its mortgage rate in stage 2 according to

$$\max_{r_b} (r_b - \psi_b^m) d_b(r_b) + \eta(\rho_b - \psi_b^w) \sum_{s \in S_b} d_s(r_b). \quad (8)$$

The first-order condition yields

$$r_b = \psi_b^m + \left(\frac{\partial \ln(d_b(r_b))}{\partial r_b} \right)^{-1} + \eta(\rho_b - \psi_b^w) \sum_{s \in S_b} \frac{\partial d_s(r_b)/\partial r_b}{\partial d_b(r_b)/\partial r_b}. \quad (9)$$

- **First two terms.** Marginal cost plus the standard markup from differentiated-product mortgage demand.
- **Third term.** An additional markup due to warehouse profits: lowering the bank's mortgage rate steals business from shadow banks and therefore reduces their warehouse borrowing.
- **Economic meaning.** This is the model counterpart of the **funding channel**. Vertically integrated banks charge higher mortgage rates because they can recover rents upstream.

3.7 Shadow bank problem

In stage 1, shadow bank s chooses the warehouse lender that maximizes anticipated mortgage profit times relationship-specific convenience:

$$\max_{b \in B} \Pi_{sb} = \pi_s^*(\rho_b) \times \xi_{bs}. \quad (10)$$

This implies choice probability

$$s_{sb}^w = \Pr(\Pi_{sb} \geq \Pi_{sb'}, \forall b'). \quad (11)$$

With type-II extreme value shocks, the closed-form share is

$$s_{sb}^w(\rho_b) = \frac{\pi_s^*(\rho_b)^\sigma}{\sum_{b'} \pi_s^*(\rho_{b'})^\sigma}. \quad (12)$$

In stage 2, after choosing a warehouse line, shadow bank s sets its mortgage rate by solving

$$\max_{r_s} \pi_s(r_s) = (r_s - \eta\rho_b - \kappa_s) d(r_s), \quad (13)$$

which implies

$$r_s = \eta\rho_b + \kappa_s + \left(\frac{\partial \ln(d_s(r_s))}{\partial r_s} \right)^{-1}. \quad (14)$$

- **Interpretation of σ .** This parameter governs how substitutable warehouse lenders are. Large σ means almost perfect substitutability; small σ means relationships matter a lot.
- **Interpretation of (13)–(14).** Shadow-bank mortgage pricing is standard differentiated-product pricing, but the marginal cost includes warehouse funding costs. So upstream prices transmit directly into downstream mortgage prices.

3.8 Household demand and equilibrium

Household i 's utility from lender option j is

$$u_{ij} = -\alpha r_j + q_j + \varepsilon_{ij}. \quad (15)$$

Integrating over the idiosyncratic preference shocks yields lender-level demand

$$d_j(r_j) = \frac{\exp(-\alpha r_j + q_j)}{\exp(v_0) + \sum_{j'} \exp(-\alpha r_{j'} + q_{j'})} F. \quad (16)$$

- **Interpretation.** Mortgage demand is a discrete-choice system with an outside option (cash purchase or rental).
- **Role of α .** α is household price sensitivity.
- **Role of q_j .** Service quality or nonprice attributes shift demand.
- **Equilibrium.** Banks set warehouse prices, shadow banks choose funding providers, lenders set mortgage rates, and households choose among mortgage products and the outside option.

3.9 Calibration and counterfactual mapping

The calibration uses MSA-year moments from conforming mortgage data and shadow-bank call reports. Several moments map directly into structural objects.

Service quality is pinned down from observed rates and market shares:

$$\tilde{q}_b = \alpha r_f + \ln \left(\frac{d_f/F}{d_0/F} \right), \quad \tilde{q}_s = \alpha r_s^f + \ln \left(\frac{d_s^f/F}{d_0/F} \right). \quad (17)$$

Shadow-bank marginal costs are backed out from pricing equations:

$$\eta \rho_f + \kappa_s = r_s^f - \frac{1}{\alpha(1 - d_s^f/F)}, \quad \eta \rho_w + \kappa_s = r_s^w - \frac{1}{\alpha(1 - d_s^w/F)}. \quad (18)$$

Banks' mortgage-origination marginal cost is

$$\psi_b^m = r_m - \frac{1}{\alpha(1 - d_m/F)}. \quad (19)$$

Consumer surplus changes are evaluated as

$$\Delta CS = \frac{1}{\alpha} \left[\ln \sum_j \exp(-\alpha r_j + q_j) - \ln \sum_j \exp(-\alpha \tilde{r}_j + q_j) \right] \times 2.78. \quad (20)$$

- **Externally calibrated parameter.** Household price sensitivity is fixed at $\alpha = 1.14$, following Buchak et al. and DeFusco-Paciorek.
- **Estimated differentiation.** The model estimates substantial lender differentiation: $\sigma \approx 0.52$ from cross-sectional calibration and $\sigma \approx 0.30$ from the oil-shock exercise.
- **Estimated warehouse duration.** $\eta = 0.09$, which maps to about a 3-month warehouse duration.
- **Cost advantage of vertically integrated banks.** The estimated marginal cost of warehouse lending is about 25% lower for vertically integrated banks than for warehouse-only banks.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- The raw fact that banks lend to shadow banks could simply reflect similarities in business models or geography, not upstream market power.
- Likewise, a correlation between a bank's warehouse activity and its mortgage rates could be driven by local demand conditions rather than by strategic behavior.
- The paper therefore needs variation that isolates *relationship formation*, *lender differentiation*, and *strategic conduct* separately.

4.2 Why overlap identifies the matching pattern

- The baseline pair-level regression includes shadow-bank-year and bank-year fixed effects, so it compares which banks a given shadow bank is more likely to borrow from *within year*, and which shadow banks a given bank is more likely to fund *within year*.
- This design absorbs time-varying bank balance-sheet conditions, aggregate shadow-bank growth, and borrower-specific or lender-specific trends.
- The remaining identifying variation comes from pair-level differences in downstream mortgage-market overlap and proximity.
- The historical-overlap robustness check strengthens the interpretation by showing that established geographic overlap predicts later warehouse relationships.

4.3 Why the oil-price shock identifies differentiation

- The 2014 oil-price collapse creates a quasi-exogenous net-worth shock to banks with high oil-and-gas exposure.
- If warehouse lenders were close substitutes, shadow banks should switch away from exposed banks once those banks raise effective funding costs.
- The paper instead finds higher funding costs but almost no switching in provider choice, only reduced usage of exposed lines.
- That response is exactly what one expects when relationships are differentiated and existing providers have market power.

4.4 Why the strategic-channel regressions are informative

- The cannibalization regression uses bank-year and state-year fixed effects with county controls, so the identifying variation is how a given bank funds shadow banks differently across counties with different downstream concentration.
- The funding-channel regression uses bank-time and MSA-time fixed effects, plus rich borrower controls at the loan level, so the variation comes from bank-MSA-quarter differences in how intensely a bank funds shadow banks.
- These designs do not by themselves prove the full structural model, but they line up very closely with the specific additional-markup terms that emerge from the theory.

4.5 What calibrates the structural parameters

- The calibration matches warehouse lending rates, warehouse market shares, mortgage rates, mortgage market shares, and outside-option shares.
- Service quality is pinned down by observed rates and market shares.
- Shadow-bank and bank marginal costs are backed out from optimal pricing formulas.

- The differentiating parameter σ is disciplined both by cross-sectional allocation patterns and by the oil-shock response.

4.6 Relation to the literature

- Relative to the shadow-banking literature, the paper shifts attention from *whether* nonbanks compete with banks to *how their funding dependence on banks changes that competition*.
- Relative to vertical-foreclosure models, the paper applies the logic to differentiated financial products and to a setting where the upstream good is a funding input.
- Relative to mortgage-demand papers, the contribution is to jointly model upstream warehouse funding and downstream mortgage origination rather than treating the shadow-bank funding side as passive or frictionless.

5 Tables and figures

5.1 Table 1 and Figure 1: Shadow-bank funding structure and who provides warehouse lines

Table 1
Summary statistics

	N	Mean	Median	SD
<i>A. Shadow banks</i>				
Asset (billions)	6,487	0.47	0.07	1.51
Mortgage (%)	6,487	68.08	76.67	24.87
Warehouse debt facility (%)	6,487	60.87	69.95	25
Mortgage origination (billions)	1,893	2.10	0.74	5.35
Number of states of origination	1,893	22.46	19	16.01
Geographic concentration	1,893	0.1	0.06	0.12
Number of credit lines	6,487	3.62	3	2.19
Credit limit (millions)	22,272	111.06	40	609.16
Lender share SD	5,874	0.14	0.12	0.11
Usage (millions)	22,272	61.55	18.11	428.36
Lender share SD	5,800	0.2	0.16	0.15
Estimated interest rate spread	5,171	0.03	0.03	0.02
Time-series SD	333	0.01	0.01	0.01
Number of shadow banks	416			
<i>B. Warehouse banks</i>				
Asset (billions)	3,067	80.93	2.39	304.23
Total loan (%)	3,067	0.68	0.69	0.12
Mortgage (%)	3,067	0.47	0.48	0.17
Mortgage origination (billions)	768	4.51	0.19	19.65
Number of states of origination	768	15.62	8.00	16.92
Geographic concentration	768	0.21	0.15	0.20
Number of banks	207			
<i>C. Bank-shadow bank pairs</i>				
Mortgage market overlap Financing	7,559	12.62	7.33	14.56
Credit limit Financing	7,559	84M	30M	662M
Mortgage market overlap No financing	264,984	5.49	2.19	8.22
Number of bank-shadow-bank pairs	272,543			
Unconditional likelihood of financing	2.77%			

This table reports summary statistics for HMDA-matched shadow banks and warehouse banks. Panel A shows shadow banks' summary statistics. Panel B shows warehouse banks' summary statistics. Panel C shows the summary statistics for all possible pairs of shadow banks and warehouse banks. The sample period in all three panels is from 2011 to 2017. Balance sheets are observed at a quarterly frequency. Mortgage origination activity is observed at an annual frequency.

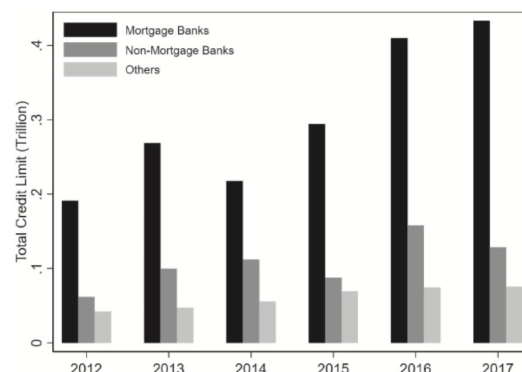


Figure 1
Upstream: Who provides warehouse lines?
This figure presents the amount of warehouse lines of credit extended by each type of lenders to shadow banks. Mortgage banks are commercial banks that originate mortgages and have a record in the HMDA. Nonmortgage banks are commercial banks that do not have a record in the HMDA. Others include the GSE, real estate investment trusts (REITs), and other financial institutions. I calculate mortgage origination market shares using HMDA data. Sources: Warehouse lending market shares are calculated using shadow bank call report filings to state regulators.

Read:

- The core descriptive fact is that shadow banks are highly mortgage-focused and heavily warehouse-funded: about 90% of their mortgage assets on balance sheet are financed with warehouse credit.
- In the matched sample there are 416 shadow banks, 207 warehouse banks, and over 22,000 warehouse lines. The unconditional probability of any given bank-shadow-bank pair having a funding relationship is only 2.77%.

- Figure 1 shows that mortgage banks—banks with in-house mortgage origination—supply about 70% of total warehouse credit extended to shadow banks.
- This figure immediately overturns the simple view that shadow banks are funded by neutral capital-market investors. The main funders are direct downstream competitors.

5.2 Figure 2 and Figure 3: Shadow banks borrow from banks they compete with

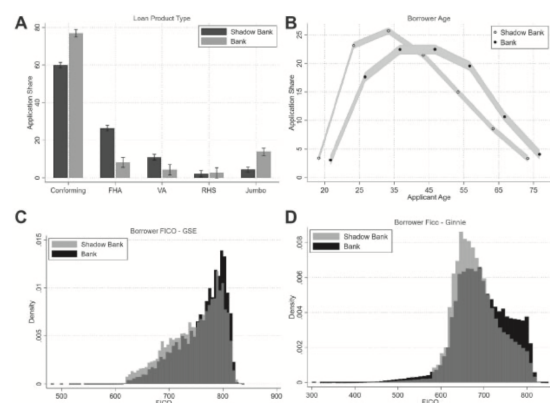


Figure 2
Downstream: Nature of competition
This figure compares the distribution of loan product type (panel A), borrower age (panel B), and borrower FICO (panels C and D) of loans originated by shadow banks and the banks that lend to shadow banks. *Sources:* Panels A and B use HMDA data from 2018 to 2019 because HMDA started collecting borrower age information since 2018. Panel C uses the Single-Family Loan Performance data disclosed by Fannie Mae and Freddie Mac from 2011 to 2017 to match with the main analysis sample period. Panel D uses the Loan Level Disclosure Data from Ginnie Mae from 2013 to 2017.

Table 3
Transmission of bank net worth shock: Shadow bank borrowing

	(1) Funding costs (pp)	(2) Funding costs (pp)	(3) Matched likelihood (%)	(4) Usage (millions)	(5) Usage share (%)
High oil shock $\times$ Post	0.608* (0.359)	0.736* (0.375)	0.012 (0.315)	-8.552* (4.882)	-2.189* (1.298)
Shadow bank controls	No	Yes	No	No	No
Shadow bank FE	Yes	Yes	No	No	No
Year FE	Yes	Yes	No	No	No
Shadow bank $\times$ Year FE	No	No	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No
Bank-shadow bank FE	No	No	No	Yes	Yes
N	1,108	1,104	67,175	15,246	15,246
R-squared	.072	.083	.116	.339	.494

This table reports the effect of banks' net worth shocks on their relationship shadow banks' borrowing. In columns 1 and 2, observations are at shadow bank-year level. The dependent variable is the interest rate spread in percentage points. *HighOilShock* is an indicator of whether the share of the shadow bank's total funding obtained from *exposed banks* is in the top quartile in 2013, where *exposed banks* are those banks whose balance sheet exposure to the O&G industry before the shock was in the top quartile. *Post* is 1 for years after the shock. Shadow bank controls include log of asset, net income to asset ratio, equity to asset ratio, operating cash flow to asset ratio. In column 3, the underlying sample includes all pairs of shadow banks and banks offering warehouse lines. The dependent variable is an indicator that equals 100 if the shadow bank receives a warehouse line from the bank in a given year. In columns 4 and 5, the underlying sample includes all warehouse lines. The dependent variable in column 4 is the usage of credit line at quarter-end in millions and in column 5 is the share of total credit limit being used. In columns 3-5, *HighOilShock* indicates whether the bank is an exposed bank defined above. *Post* equals one for years after the shock. In all columns, I use two years of observations before and after the event date. Standard errors are reported in the parentheses, clustered by shadow banks and banks. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read:

- Figure 2 shows that shadow banks and their funding providers overlap substantially in downstream market segments, even if they differ somewhat in product mix and borrower composition.
- Figure 3 shows that about 80% of warehouse banks lend only to shadow banks in their own mortgage markets, even though more than 70% had the option to lend outside those markets.
- More than 80% of shadow banks also receive all funding from banks in their own mortgage markets.
- The message is not merely that banks fund shadow banks. It is that they fund *local competitors* in the same downstream business.

5.3 Table 2: Matching in the warehouse lending market

Read:

- In the saturated pair-level specification, a one-standard-deviation increase in mortgage-market overlap raises the likelihood of a funding relationship by 1.88 percentage points.
- Relative to the unconditional financing probability of 2.77%, this is very large—roughly 68% of the baseline likelihood.
- The finding is robust when the choice set is restricted to nearby lenders, lenders with similar geographic concentration, and lenders in similar size buckets.

- Historical geographic overlap from 2008–2010 also predicts later funding ties, reinforcing the interpretation that established downstream presence shapes upstream matching.

Table 2
Matching in the warehouse lending market

A. Baseline							
	(1) Full sample	(2) Full sample	(3) Nearby lenders	(4) Regional lenders	(5) National lenders	(6) Small lenders	(7) Big lenders
Overlap	1.47*** (0.383)	1.88*** (0.429)	2.75*** (0.507)	0.99*** (0.313)	2.64*** (0.667)	0.97*** (0.311)	2.18*** (0.604)
log(HQ distance)	-0.81*** (0.156)	-0.85*** (0.168)	-1.00*** (0.185)	-1.17*** (0.214)	-0.81*** (0.227)	-1.14*** (0.181)	-0.77*** (0.305)
Bank controls	Yes	No	No	No	No	No	No
Shadow bank×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank×Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
N	272,237	272,346	69,649	102,426	34,429	102,446	34,172
R-squared	.06	.12	.15	.12	.16	.11	.16

B. Historical geographic distribution							
Full sample				Nearby lenders			
	(1) Overlap	(2) Warehouse lending	(3) Warehouse lending	(4) Overlap	(5) Warehouse lending	(6) Warehouse lending	
Overlap 2008–2010	0.58*** (0.039)		1.55*** (0.509)	0.61*** (0.028)		2.52*** (0.460)	
Overlap		2.66*** (0.763)			4.16*** (0.692)		
log(HQ distance)	-0.03** (0.013)	-0.91*** (0.238)	-0.99*** (0.246)	0.00 (0.009)	-1.05*** (0.227)	-1.04*** (0.238)	
Shadow Bank×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	167,100	167,100	167,100	42,580	42,580	42,580	
R-squared	.70	.02	.15	.67	.03	.18	

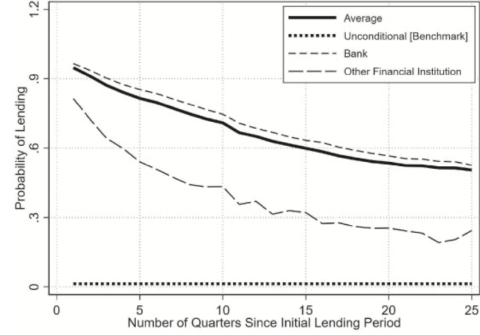


Figure 4
Upstream: Persistent warehouse lending relationships
This figure shows the likelihood of having a warehouse lending relationship in quarters 1 to 25, conditional on having a warehouse lending relationship in quarter 0. The underlying sample is a panel data set that contains all possible pairs between warehouse banks and shadow banks. A warehouse lender is included in the data set if it conducts warehouse lending business and lends to at least one shadow bank in a particular quarter. To find the conditional probability, I find the first quarter of every warehouse lending relationship and define it as quarter 0 and then track these matched bank-shadow bank pairs for 25 quarters. The conditional probability in quarter t is calculated as the share of bank-shadow bank pairs that had a warehouse lending relationship in quarter 0 and still have a warehouse lending relationship after t quarters. As a benchmark, the dotted line plots the unconditional probability of having a lending relationship in each quarter. The unconditional probability in quarter t is calculated as the total share of bank-shadow bank pairs that have a warehouse lending relationship in quarter t , regardless of whether they had a warehouse lending relationship in quarter 0.

5.4 Figure 4, Table 3, and Figure 5: Persistent relationships and the oil-shock test

Read:

- Figure 4 shows very persistent funding relationships: conditional on borrowing from a bank today, the probability of borrowing from the same bank next year is close to 90%, and it is still above 50% after five years.
- Table 3 shows that after the oil-price shock, treated shadow banks' funding costs rise by more than 60 bps relative to untreated peers.
- Yet the probability of switching lenders does not move significantly; instead, treated shadow banks reduce usage of exposed lines by about \$8.6 million, or about 2% of the credit limit.
- Figure 5 confirms these results dynamically and shows no pretrend. This is the paper's cleanest reduced-form evidence that warehouse lenders are differentiated.

5.5 Table 4: The cannibalization channel

Read:

- The key coefficient is negative: a one-standard-deviation rise in downstream HHI is associated with a 58-bps decline in the share of shadow banks financed by a local bank.
- The result holds across regional, national, small, and big lender subsamples.
- This is exactly what the cannibalization logic predicts: where downstream mortgage profits are more valuable, local banks pull back from funding local shadow-bank competitors.

Table 3

Transmission of bank net worth shock: Shadow bank borrowing

	(1) Funding costs (pp)	(2) Funding costs (pp)	(3) Matched likelihood (%)	(4) Usage (millions)	(5) Usage share (%)
High oil shock $\times$ Post	0.608* (0.359)	0.736* (0.375)	0.012 (0.315)	-8.552* (4.882)	-2.189* (1.298)
Shadow bank controls	No	Yes	No	No	No
Shadow bank FE	Yes	Yes	No	No	No
Year FE	Yes	Yes	No	No	No
Shadow bank $\times$ Year FE	No	No	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No
Bank-shadow bank FE	No	No	No	Yes	Yes
N	1,108	1,104	67,175	15,246	15,246
R-squared	.072	.083	.116	.339	.494

This table reports the effect of banks' net worth shocks on their relationship shadow banks' borrowing. In columns 1 and 2, observations are at shadow bank-year level. The dependent variable is the interest rate spread in percentage points. *HighOilShock* is an indicator of whether the share of the shadow bank's total funding obtained from *exposed banks* is in the top quartile in 2013, where *exposed banks* are those banks whose balance sheet exposure to the O&G industry before the shock was in the top quartile. *Post* is 1 for years after the shock. Shadow bank controls include log of asset, net income to asset ratio, equity to asset ratio, operating cash flow to asset ratio. In column 3, the underlying sample includes all pairs of shadow banks and banks offering warehouse lines. The dependent variable is an indicator that equals 100 if the shadow bank receives a warehouse line from the bank in a given year. In columns 4 and 5, the underlying sample includes all warehouse lines. The dependent variable in column 4 is the usage of credit line at quarter-end in millions and in column 5 is the share of total credit limit being used. In columns 3–5, *HighOilShock* indicates whether the bank is an exposed bank defined above. *Post* equals one for years after the shock. In all columns, I use two years of observations before and after the event date. Standard errors are reported in the parentheses, clustered by shadow banks and banks. * $p < .1$; ** $p < .05$; *** $p < .01$.

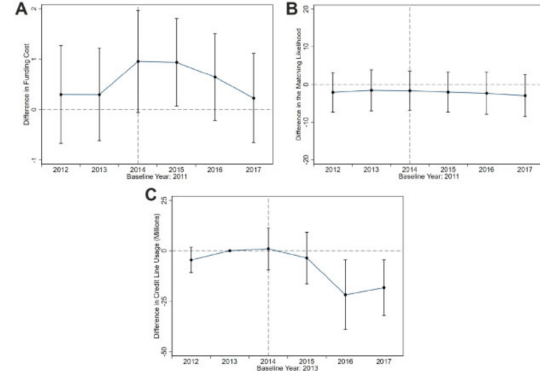


Figure 5

Transmission of bank net worth shock

This figure plots the estimated coefficients of dynamic difference-in-differences regressions. Panel A corresponds to column 1 of Table 3. Panel B corresponds to column 3 of Table 3. Panel C corresponds to column 4 of Table 3. The vertical bars represent 90% confidence intervals. Standard errors are double clustered by shadow bank and bank.

- The table is important because it shows strategic conduct already in the reduced form, before any structural calibration.

Table 4

The cannibalization channel

	# Shadow banks financed/total # shadow banks (%)				
	(1) Full sample	(2) Regional lenders	(3) National lenders	(4) Small lenders	(5) Big lenders
Mortgage market HHI	-0.58*** (0.208)	-0.49** (0.200)	-0.91*** (0.177)	-0.20** (0.094)	-0.98** (0.111)
County controls	Yes	Yes	Yes	Yes	Yes
Bank $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
R-squared	.78	.78	.54	.75	.38
N	227,754	168,398	59,354	171,718	56,035

This table provides evidence for the cannibalization channel, which states that banks are less likely to finance shadow banks in more concentrated downstream markets. Downstream concentration is measured by mortgage market Herfindahl-Hirschman index, which is scaled by sample standard deviation. The underlying sample contains bank-county-year level observations. I restrict to counties where banks have mortgage origination, which maps to vertically integrated banks in the model. The outcome variable is the share of shadow banks in the county that are financed by the bank, expressed in percentage points, which maps to s_{it}^B in the model. Column 1 uses the full sample. Columns 2–5 use subsamples of regional banks, national banks, small banks, and big banks, respectively, which are defined in the same way as in Section 2. County controls include log income per capita, log population, and employment rate. Standard errors are double-clustered at county and bank levels, reported in the parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5

The funding channel

	(1) Loan level	(2) Bank level
Warehouse lending market share	0.19*** (0.02)	0.12*** (0.04)
Borrower controls	Yes	No
Bank $\times$ Time FE	Yes	Yes
MSA $\times$ Time FE	Yes	Yes
Observations	1,171,525	36,955
R-squared	.40	.41

This table shows how banks' mortgage pricing is associated with their warehouse lending activity. Column 1 reports results at loan-level, in which the dependent variable is mortgage interest rates. Column 2 reports results at institution-level, in which the dependent variable is weighted average residualized mortgage interest rates. Residualized mortgage interest rates (i.e., rate premiums) are found by residualizing mortgage interest rates using LTV, DTI, FICO, the squared terms, and their interactions with origination quarters. Standard errors in parentheses are clustered at the MSA-quarter level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 5: The funding channel

Read:

- A one-percentage-point increase in a bank's warehouse lending market share is associated with about a 19-bps increase in the mortgage rates it charges on individual loans.
- The same pattern appears in bank-level rate-premium regressions using bank-time and MSA-time fixed effects.
- The interpretation is that banks soften downstream price competition when they earn more upstream profit from the shadow banks they finance.
- This table is the reduced-form counterpart to the additional downstream markup in equation (9).

5.7 Table 6: Calibration moments and parameters

Read:

- The model is disciplined by both upstream and downstream moments: warehouse lending rates, warehouse lending market shares, mortgage rates, mortgage market shares, and the outside-option share.
- The calibrated differentiation parameter is substantial: $\sigma \approx 0.52$ in the cross-sectional calibration and $\sigma \approx 0.30$ from the oil-shock exercise.
- The calibrated warehouse duration is $\eta = 0.09$, which corresponds to roughly three months.
- Vertically integrated banks are estimated to have lower warehouse marginal cost than warehouse-only banks ($\psi_f^w = 3.28$ versus $\psi_w^w = 4.38$), which is the model's way of capturing the synergy between upstream and downstream lending.

Table 6
Model moments and parameters

A. Moments targeted in calibration and fit		
Moment	Data	Model
Warehouse lending rate		
Vertically integrated banks (ρ_f)	3.69	3.67
Warehouse banks (ρ_w)	4.38	4.38
Warehouse lending market share		
Vertically integrated banks ($s_f^w N_f$)	84%	89%
Warehouse banks (s_w^w)	16%	11%
Mortgage rate		
Vertically integrated banks (r_f)	4.4	4.401
Mortgage banks (r_m)	4.25	4.400
Shadow banks financed by vertically integrated banks (r_s^f)	4.45	4.43
Shadow banks financed by warehouse banks (r_s^w)	4.52	4.49
Mortgage market share		
Vertically integrated banks ($\frac{d_f^f}{N_f}$)	20%	19%
Mortgage banks ($\frac{d_m^f}{N_m}$)	18%	25%
Shadow banks financed by vertically integrated banks ($\frac{d_s^f}{N_f}$)	24%	23%
Shadow banks financed by nonvertically integrated ($\frac{d_s^w}{N_f}$)	5%	3%
Outside option ($\frac{d_o}{N_f}$)	33%	30%
B. Calibrated parameters		
Parameter	Description	Value
σ	Product differentiation (cross-sectional variation)	0.52 (0.17)

Table 7
Counterfactual: Banks' upstream market power

	Δ SB funding cost	Δ SB mortgage rate	Δ SB origination volume	Δ Total credit supply	Δ Consumer surplus
Average MSA					
Level change	+46.4 bps	+4.1 bps	−49.4M	−21.0M	−10.0M
Percentage change	+12.4%	+0.9%	−3.4%	−0.5%	(−0.18 cents per dollar)
Least concentrated					
Level change	+37.1 bps	+3.3 bps	−123.6M	−47.9M	−83.6M
Percentage change	+11.3%	+0.8%	−2.3%	−0.4%	(−0.58 cents per dollar)
Most concentrated					
Level change	+61.8 bps	+5.5 bps	−12.5M	−6.6M	−1.9M
Percentage change	+18.8%	+1.2%	−5.1%	−0.8%	(−0.13 cents per dollar)
Least preference for SB credit					
Level change	+28.4 bps	+2.5 bps	−10.1M	−4.3M	−4.9M
Percentage change	+8.7%	+0.6%	−2.5%	−0.2%	(−0.15 cents per dollar)
Most preference for SB credit					
Level change	+62.8 bps	+5.6 bps	−39.1M	−16.6M	−5.4M
Percentage change	+19.1%	+1.3%	−3.8%	−0.84%	(−0.21 cents per dollar)

This table shows the quantitative importance of banks' upstream market power. I compare the baseline model to a counterfactual world where banks do not have upstream market power by making σ sufficiently large and thus making banking relationships perfectly substitutable. Column 1 reports changes in shadow banks' funding costs. Column 2 reports changes in shadow banks' mortgage rate. Column 3 reports changes in shadow banks' origination volume. Column 4 reports changes in total credit supply. Column 5 reports changes in consumer surplus.

5.8 Table 7: Counterfactual impact of banks' upstream market power

Read:

- In the average MSA, banks' upstream market power raises shadow banks' funding costs by 46.4 bps, or 12.4%, and raises their mortgage rates by 4.1 bps.
- Shadow-bank origination falls by \$49.4 million, total mortgage credit supply falls by \$21 million, and consumer surplus declines by about \$10 million in an average MSA-year.
- On a per-borrower basis, the paper maps this to about a \$638 reduction in consumer surplus for the average mortgage borrower.
- The effects are markedly stronger in more concentrated downstream markets and in regions where households value shadow-bank services more.

5.9 Figure 6: Decomposing the strategic channels

Read:

- Panels A–C isolate the cannibalization channel by shutting down the incentive to protect downstream profits in warehouse pricing.

- The cannibalization channel raises shadow banks' funding costs by about 0.9 bps and their mortgage rates by about 0.08 bps, and it is much stronger in concentrated downstream markets.
- Panel D isolates the funding channel by shutting down the incentive to account for warehouse profits in mortgage pricing. This channel raises banks' mortgage rates by about 0.8 bps and is stronger in more competitive downstream markets.
- Quantitatively, these channel decompositions are small relative to the total effect of upstream market power, but they show clearly how vertically integrated intermediaries strategically distort both markets.

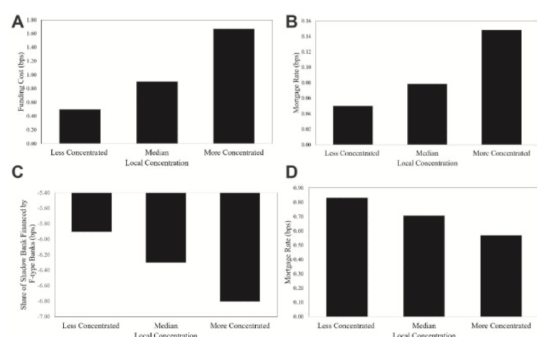


Figure 6
The economic channels
This figure shows the cannibalization channel (panels A-C) and the funding channel (panel D). In panels A-C, I compare the baseline model to a counterfactual world where vertically integrated (VI) banks do not price in the additional markup in their upstream decisions (Equation (7)). The magnitudes show the quantitative importance of the cannibalization channel for shadow banks' funding costs in panel A, for mortgage rate charged by shadow banks in panel B, and for the share of shadow banks financed by vertically integrated banks in panel C. In panel D, I compare the baseline model to a counterfactual world where vertically integrated banks do not price in the additional markup in their downstream pricing decisions (Equation (9)). The magnitudes show the quantitative importance of the cannibalization channel for banks' mortgage rate in panel D. In each panel, I show three scenarios: less concentrated markets (bottom decile), average markets, and more concentrated markets (top decile), which are defined based on the inverse of MSA-level number of mortgage lenders (i.e., $1/N$).

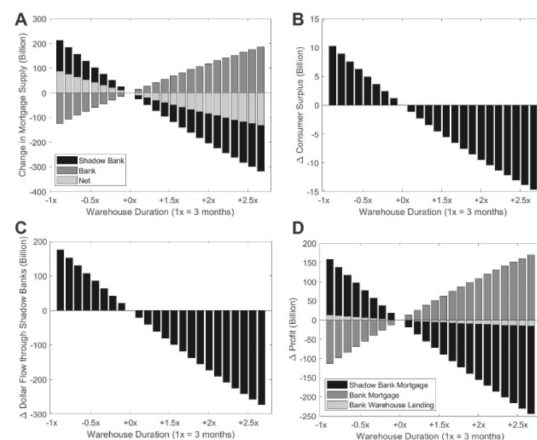


Figure 7
Counterfactual: Warehouse duration
This figure plots the impact of changing the warehouse duration η . Panel A shows changes in lending volume. To get total dollar volume, I set total market size F to be the average loan size (\$354,500) times the population in need of a mortgage (60% homeownership rate, 2 people per household, 10% of households looking for a house in a year, that is, $354,500 \times \text{Population} \times 60\% \times 10\% / 2$). Panel B shows changes in consumer surplus. Consumer surplus is calculated as a lifetime present-value dollar equivalent measure of expected utility, which is integrated over consumer-specific preference shocks, while assuming a subjective discount rate of 4% over a period of 3 years. Panel C shows dollar flow through shadow banks via warehouse lending. Panel D shows intermediary profit.

5.10 Figure 7: Secondary-market innovation and warehouse duration

Read:

- The figure studies what happens when warehouse duration becomes shorter, as would happen with faster loan sale, eMortgage adoption, or GSE process improvements.
- A 45-day shorter time-to-sale expands shadow-bank mortgage lending by about \$150 billion and crowds out roughly \$50 billion of bank lending.
- Consumer surplus increases by nearly \$50 billion, while shadow-bank profits rise by about \$75 billion.
- The key message is that reducing shadow banks' dependence on competitor funding can materially increase downstream competition.

6 Short summary

- **Main message.** Shadow banks are not funded by a neutral capital market; they are largely funded by competing banks, and that funding structure matters for mortgage-market competition.
- **Main mechanism.** Banks have upstream market power in warehouse lending because those relationships are differentiated. That market power softens downstream competition through a cannibalization channel and a funding channel.
- **Main empirical lesson.** Upstream funding frictions meaningfully restrict shadow-bank lending and harm borrowers, especially in concentrated markets and in places where households rely more on shadow-bank services.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that shadow banks' funding and banks' mortgage competition cannot be studied separately.
- Banks that fund shadow banks are often the same institutions competing with them downstream, and this vertical link creates strategic distortions.
- Warehouse lending relationships are differentiated enough that banks can charge markups and use upstream pricing to influence downstream competition.
- In the calibrated model, this upstream market power raises shadow-bank funding costs, raises mortgage rates, reduces credit supply, and lowers consumer surplus.

7.2 Economic intuition

- Shadow banks look like aggressive competitors because they do not have deposits and are often less regulated. But their business model makes them dependent on warehouse funding.
- If warehouse lenders are their downstream competitors, those competitors can use funding prices strategically.
- A vertically integrated bank then earns profits in both markets and chooses prices to maximize *total* profit, not just the profit in one market at a time.
- This means less aggressive funding of competitors upstream and less aggressive mortgage pricing downstream. Competition is softened from both directions.

7.3 Takeaway

The paper's broad lesson is that the rise of shadow banking does not automatically imply stronger competition. It depends on who controls the inputs shadow banks need to operate. In the mortgage market, banks fund the very rivals that compete with them, and this allows upstream market power to spill into downstream prices and quantities. The result is especially important for policy: reforms

that speed up loan sales or reduce shadow banks' dependence on competitor funding can improve mortgage-market competition without directly regulating mortgage demand. In that sense, Jiang's paper is best read as a vertical-structure paper about household credit markets.